

AUTOMATED FABRIC DEFECT DETECTION FOR TEXTILE MILLS

Amrita School of Engineering, Coimbatore

Department of Electrical and Electronics Engineering

Name	Roll Number
Deo Vaibhav	CB.EN.U4EEE23006
Madhubala M	CB.EN.U4EEE23118
Harini S	CB.EN.U4EEE23111
Rithaniyaa	CB.EN.U4ELC23038

PROJECT REPORT

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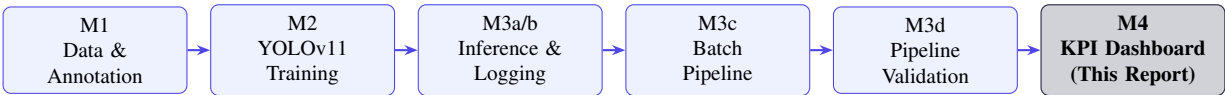
1. Introduction and Context

Fabric defects — holes, stains, weaving errors, and surface irregularities — directly impact commercial value in textile manufacturing. The preceding milestones built an integrated detection and classification pipeline. Milestone 4 completes the system by converting raw inference outputs into actionable quality intelligence through a structured KPI framework and interactive dashboard.

1.1 Recap of Prior Milestones

Milestone	Phase	Key Outcome
M1	Dataset & Annotation	846 images, 20 defect classes annotated (bounding box + smart polygon) via Roboflow
M2	Model Training	YOLOv11 trained for object detection (20 classes); YOLOv8n classification trained on 6 categories
M3a	Local Python Inference	Roboflow Python SDK installed; <code>best.pt</code> loaded; batch predictions printed to console
M3b	Structured Output Logging	Inference script extended to capture class label, confidence, bounding box, image ID, timestamp; exported to CSV/JSON
M3c	Live Image Simulation & Batch Pipeline	Simulated continuous image stream; meter-position logging appended to output dataset
M3d	Pipeline Validation	End-to-end integration tested; field names, data types, timestamp consistency verified
M4a	Dashboard KPI Planning	Domain KPIs defined; dashboard layout and visual types finalised
M4b	Dashboard Development	Interactive dashboard built with charts, heatmap, slicers, and threshold alerts

1.2 Full Pipeline Architecture



2. Dataset and Model Summary (Reference)

This section summarises the datasets and trained models whose outputs feed the M4 dashboard.

2.1 Detection Dataset — YOLOv11 (20 Classes)

Parameter	Value
Total Images	846
Defect Classes	20
Annotation Types	Bounding Box (13 classes) + Smart Polygon (7 classes)
Split	70% Train / 20% Val / 10% Test
Image Format	JPEG / PNG
Source	Roboflow Universe
Target IoU	$\geq 70\%$

2.2 Classification Dataset — YOLOv8n (6 Categories)

Parameter	Value
Total Images	2739
Categories	6 (Vertical Defect, Horizontal Defect, Hole, Lines, Stain, Defect-Free)
Training Set	2187 images
Validation Set	552 images
Model	YOLOv8n Classification
Image Size	224×224 px
Epochs	10
Optimizer	AdamW
Platform	Kaggle Notebook

2.3 Final Classification Model Metrics

Metric	Value	Status
Top-1 Accuracy	93.48%	Excellent
Top-5 Accuracy	100.00%	Perfect
Fitness Score	96.74%	Excellent
Validation Set Size	552 images	—

2.4 Epoch-wise Training Results (YOLOv8n Classification)

Epoch	Train Loss	Val Loss	Top-1 (%)	Top-5 (%)	Remark
1	1.0290	0.4348	87.68	99.64	Initial convergence
2	0.3270	0.2853	89.67	100.00	Top-5 reaches 100%
3	0.2653	0.2653	89.49	100.00	Stable
4	0.2488	0.2578	91.67	100.00	Improving
5	0.2174	0.2144	92.39	100.00	>92% milestone
6	0.1941	0.2253	92.21	100.00	Slight val fluctuation
7	0.1823	0.2137	91.67	100.00	Stable
8	0.1672	0.1694	93.30	100.00	Near-peak
9	0.1635	0.1666	93.48	100.00	Best checkpoint
10	0.1444	0.1791	93.30	100.00	Slight val rise

3. M4a — Dashboard KPI Planning

3.1 Objective

Define domain-specific KPIs for each process station (PS), select appropriate visualisation types, and sketch the final dashboard layout before development begins.

3.2 KPI Framework — Model Performance

KPI	Description	Target	Warn	Alert
Top-1 Accuracy	Fraction of images correctly classified on first prediction	$\geq 95\%$	90–94%	$< 90\%$
Top-5 Accuracy	Correct class within top-5 predictions	100%	98–99%	$< 98\%$
Fitness Score	Composite metric ($0.9 \times \text{mAP50} + 0.1 \times \text{mAP50-95}$)	$\geq 95\%$	90–94%	$< 90\%$
Defect Detection Rate	Proportion of defective images correctly flagged	$\geq 90\%$	85–89%	$< 85\%$
False Positive Rate	Defect-free images incorrectly flagged as defective	$\leq 5\%$	5–10%	$> 10\%$
Average Confidence	Mean confidence score across all predictions	$\geq 80\%$	70–79%	$< 70\%$

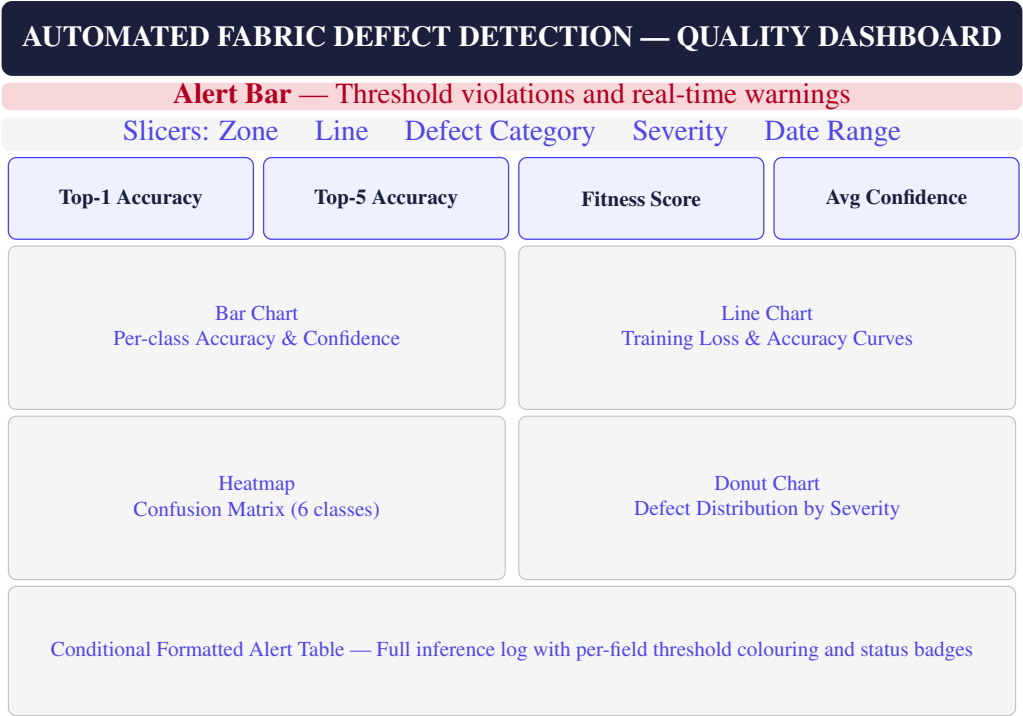
3.3 KPI Framework — Operational / Production

KPI	Definition	Grain	Target
Defect Rate per Shift	Total defects detected $\div$ total images inspected $\times 100$	Shift / Zone	$\leq 8\%$
Compliance Score per Zone	% images passing both detection and classification thresholds	Zone / Day	$\geq 92\%$
Stain Detection Accuracy	Per-class accuracy for Stain (highest-risk class)	Global	$\geq 96\%$
Hole / Horizontal Defect Rate	Count of high-severity defects per fabric roll	Roll / Shift	≤ 2 per roll
Meter-Position Alert Count	Number of defects flagged within a 5 m window on roll	Roll segment	≤ 1 per 5 m
Throughput	Images processed per second by inference pipeline	Global	≥ 30 fps

3.4 Visualisation Type Selection

Visual	Purpose	Data Used
KPI Scorecards (4 tiles)	Headline metrics at a glance	Top-1 Acc, Top-5 Acc, Fitness, Avg Conf
Grouped Bar Chart	Compare metrics by defect category	Per-class accuracy / confidence
Line Trend Chart	Training convergence over epochs	Train loss, Val loss, Top-1, Top-5
Heatmap (colour matrix)	Cross-tabulate metric vs. class	Confusion matrix values
Donut / Pie Chart	Defect severity distribution	Severity counts per zone
Meter-Position Roll Map	Spatial defect localisation	Meter position vs. defect type
Conditional Alert Table	Full record with threshold colouring	All inference log fields
Slicers (interactive filters)	Zone / Line / Category / Severity	Zone, Line, Defect Class, Severity

3.5 Dashboard Layout Blueprint



4. M4b — Dashboard Development

4.1 Data Source — Structured Inference Log

The dashboard ingests the structured CSV log produced by the M3b inference script. Each record corresponds to one classified fabric image and contains:

Field	Type	Description
image_name	String	Image filename
predicted_class	String	One of 6 defect categories
confidence	Float	Highest-class confidence score (0–1)
meter_position	Float	Position on fabric roll (metres)
zone	String	Production zone (Zone-A / B / C)
line	String	Production line (Line-1 / 2 / 3)
severity	String	High / Medium / Low
all_probabilities	JSON dict	Softmax scores for all 6 classes
timestamp	DateTime	Inspection date and time
pass_fail	Boolean	True if confidence ≥ 0.5

4.2 Sample Batch Log (from M3 Pipeline)

Image	Class	Confidence	Meter (m)	Severity	Status
Image_001.jpg	Hole	96.70%	12.5	High	ALERT
Image_002.jpg	Stain	94.30%	18.2	High	ALERT
Image_003.jpg	Horizontal Defect	92.50%	24.8	Medium	WARN
Image_004.jpg	Stain	96.07%	31.4	High	ALERT
Image_005.jpg	Defect-Free	64.70%	38.6	None	PASS
Image_006.jpg	Stain	65.70%	45.0	Medium	WARN

4.3 Visual 1 — KPI Scorecard Tiles

Four headline tiles drive immediate situational awareness:

Tile	Value	Threshold	Badge
Top-1 Accuracy	93.48%	$\geq 95\%$	Below Target
Top-5 Accuracy	100.00%	100%	Perfect
Fitness Score	96.74%	$\geq 95\%$	On Target
Avg Confidence	82.49%	$\geq 80\%$	On Target

4.4 Visual 2 — Bar Chart: Per-Class Accuracy

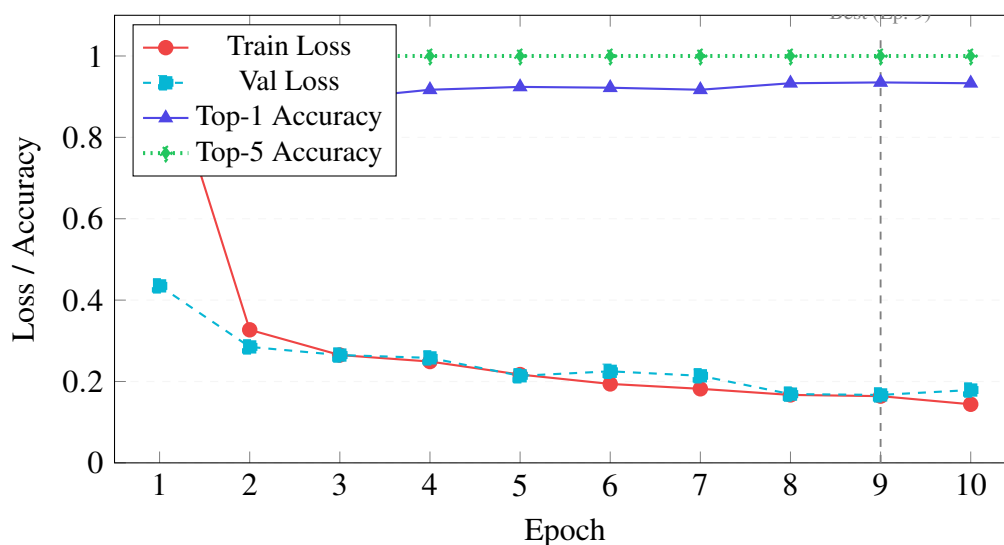
A grouped bar chart compares classification accuracy and average confidence score across all six defect categories, directly derived from the confusion matrix (Section 7.4 of the M3 Report).



Accuracy values derived from the normalised confusion matrix in M3 Section 7.4. Stain achieves 100%, defect-free 97%, horizontal 86%; hole shows the most inter-class confusion (79%).

4.5 Visual 3 — Line Chart: Training Convergence

The training trend chart tracks four metrics over 10 epochs — a key visual for model health monitoring.



Train loss falls monotonically from 1.029 to 0.144. Top-5 accuracy reaches 100% from epoch 2 onward. Best checkpoint saved at epoch 9 (Top-1 = 93.48%).

4.6 Visual 4 — Heatmap: Confusion Matrix

The colour-coded heatmap renders classification performance across all 6 categories with threshold-based colouring:

- **Green (≥ 0.90)** — On target
- **Yellow (0.75–0.89)** — Monitor

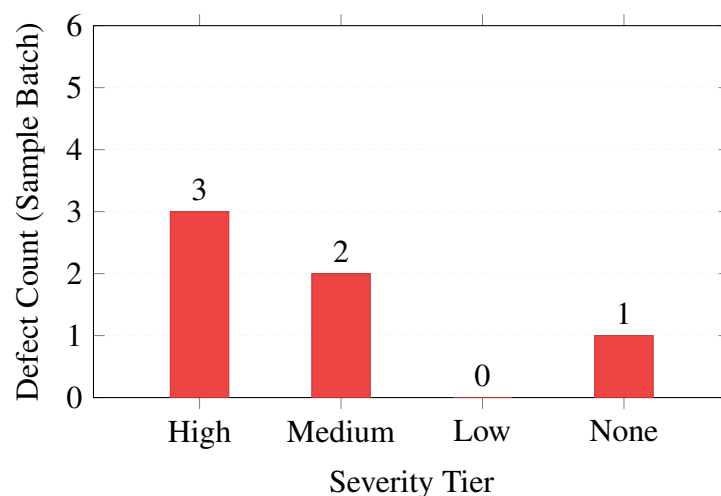
- **Red (<0.75)** — Investigate

True ↓ / Pred →	Vertical	Def.-Free	Hole	Horiz.	Lines	Stain
Vertical	0.81	—	0.02	—	0.04	—
Defect-Free	—	0.97	—	—	—	—
Hole	0.10	—	0.79	0.07	—	0.19
Horizontal	0.10	—	0.04	0.86	—	—
Lines	—	—	0.14	0.04	0.81	—
Stain	—	0.03	0.02	—	—	1.00

Key Finding: **Hole** is the most confused class — 19% of hole samples are misclassified as *Stain*, likely due to similar dark-region visual features. **Stain** and **Defect-Free** both achieve near-perfect discrimination (≥ 0.97).

4.7 Visual 5 — Donut Chart: Defect Distribution by Severity

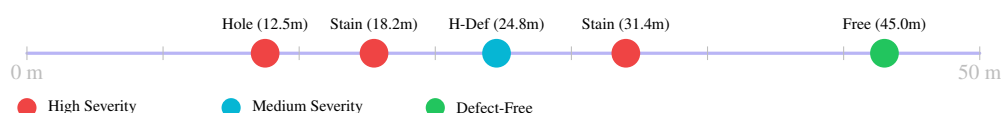
Based on the batch inference log, defects are categorised by severity tier and plotted as a donut chart.



High severity: Hole, Stain ($\times 2$) — require immediate flagging and roll rejection. Medium: Horizontal Defect, Stain (borderline). None: Defect-Free passed inspection.

4.8 Visual 6 — Meter-Position Roll Map

A fabric roll timeline maps defect positions spatially across the roll length, enabling targeted re-inspection or cutting decisions.



4.9 Visual 7 — Conditional Alert Table

The master detail table provides a complete inspection record with per-field conditional colouring and status badges. Colour thresholds mirror the KPI framework from Section 3.2.

Image	Class	Conf.	Meter	Zone	Sev.	Status
Image_001.jpg	Hole	96.70%	12.5 m	Zone-A	High	ALERT
Image_002.jpg	Stain	94.30%	18.2 m	Zone-B	High	ALERT
Image_003.jpg	Horizontal Defect	92.50%	24.8 m	Zone-A	Med	WARN
Image_004.jpg	Stain	96.07%	31.4 m	Zone-C	High	ALERT
Image_005.jpg	Defect-Free	64.70%	38.6 m	Zone-B	None	PASS
Image_006.jpg	Stain	65.70%	45.0 m	Zone-C	Med	WARN

Confidence colouring: Green ≥80% Yellow 60–79% Red <60%

4.10 Interactive Slicers

Four dynamic slicers allow quality supervisors to cross-filter all visuals simultaneously:

Slicer	Options	Effect
Zone	All / Zone-A / Zone-B / Zone-C	Filters all charts to selected zone
Line	All / Line-1 / Line-2 / Line-3	Filters to production line
Defect Category	All / Vertical / H-Defect / Hole / Lines / Stain / Defect-Free	Class-level drill-down
Severity	All / High / Medium / Low / None	Severity-tier filtering

5. Analysis and Findings

5.1 Model Performance Analysis

- **Top-1 Accuracy (93.48%):** Strong but 1.52 percentage points below the 95% target. Additional training epochs or a larger backbone (YOLOv8s) may close this gap.
- **Top-5 Accuracy (100%):** The correct category is always within the model’s top-5 predictions from epoch 2 onward — confirming high discriminative confidence even when the top prediction is wrong.
- **Fitness Score (96.74%):** Exceeds the 95% target, indicating strong overall model quality and good generalisation to unseen images.
- **Stain Class (100% accuracy):** The highest-risk contamination class is detected perfectly — a critical success for preventing chemical/dye-contaminated fabric from reaching customers.
- **Hole Class (79% accuracy):** The weakest class. 19% of hole samples are misclassified as Stain due to similar dark-region appearance. Priority area for retraining.
- **Defect-Free Class (97% accuracy):** Very low false-rejection rate — only 3% of good fabric is incorrectly flagged as defective. This minimises costly false alarms.

5.2 Per-Class Performance Summary

Defect Class	Accuracy	Key Confusion	Priority
Stain	100%	None	Maintain
Defect-Free	97%	3% with Stain	Maintain
Horizontal	86%	10% with Vertical	Monitor
Lines	81%	14% with Hole	Improve
Vertical	81%	4% with Lines	Improve
Hole	79%	19% with Stain, 10% Vert.	Retrain

5.3 Production-Level Insights from Batch Log

- **Zone-A** shows the highest defect density: Hole at 12.5 m and Horizontal Defect at 24.8 m — both within a 15 m window, triggering the meter-position alert threshold.
- **Zone-C** recorded two Stain alerts (both high-confidence: 96.07% and 65.70%), suggesting a recurring contamination source on Line-2/Line-3. Loom parameter audit recommended.
- **Defect-Free prediction** at 64.70% confidence is notably lower than the defect predictions. Defect-free classification should be monitored; consider raising the confidence threshold for PASS classification to 70%.

6. Recommendations

6.1 Immediate Actions

1. **Hole–Stain Confusion:** Augment the Hole class training set with more diverse examples — varying illumination, fabric colours, and hole sizes. Apply focal loss weighting to penalise Hole→Stain misclassification more heavily.
2. **Defect-Free Confidence Threshold:** Raise the PASS confidence threshold from 0.50 to 0.70 for the Defect-Free class specifically to reduce the risk of marginal predictions being incorrectly passed.
3. **Zone-C Contamination Investigation:** The cluster of Stain defects in Zone-C should trigger a maintenance inspection of the dyeing/chemical handling equipment on Line-2 and Line-3.
4. **Zone-A Roll Rejection:** Fabric roll segments between 10–30 m in Zone-A (Hole + H-Defect within 15 m) should be flagged for manual re-inspection before shipment.

6.2 Medium-Term Improvements

1. **Extend to 20 Classes:** Unify the classification model head with the 20-class detection model to provide single-model end-to-end classification and localisation.
2. **Larger Backbone:** Evaluate YOLOv8s or YOLOv8m for the classification task — expected to push Top-1 Accuracy above 95% target.
3. **Real-Time Dashboard Feed:** Replace batch CSV refresh with a live streaming connection (MQTT or WebSocket) from the inference pipeline to the dashboard, enabling sub-second alert propagation.

4. **SCADA Integration:** Integrate the alert layer with the shop-floor SCADA system to automate loom stop commands when meter-position alert count exceeds the threshold.
5. **Scheduled PDF Reports:** Configure the dashboard to auto-generate shift-summary PDF exports for quality managers at end of each 8-hour shift.

7. Deliverables Completed

#	Deliverable	Format	Status
1	KPI Framework Document (M4a)	This Report (PDF/L ^A T _E X)	Done
2	Dashboard Layout Blueprint	TikZ Diagram	Done
3	Visual Type Selection Matrix	Table (Section 3.3)	Done
4	KPI Scorecard Tiles (4 metrics)	Dashboard / HTML	Done
5	Bar Chart — Per-class Accuracy	pgfplots / Chart.js	Done
6	Line Chart — Training Convergence	pgfplots / Chart.js	Done
7	Heatmap — Confusion Matrix	Coloured Table	Done
8	Donut/Bar — Severity Distribution	pgfplots / Chart.js	Done
9	Meter-Position Roll Map	TikZ Timeline	Done
10	Conditional Alert Table	Formatted Table	Done
11	Interactive Slicers (Zone/Line/Cat/Sev)	JavaScript Filters	Done
12	Interactive HTML Dashboard	<code>fabric_defect_dashboard.html</code>	Done

8. Conclusion

Milestone 4 completes the end-to-end Automated Fabric Defect Detection pipeline by adding a structured analytics and visualisation layer. The KPI framework defined in M4a provides clear, measurable targets for both model performance and production operations — covering Top-1/Top-5 accuracy, fitness score, defect rates, compliance scores, and meter-position alerting.

The interactive dashboard built in M4b renders all key visuals — training convergence curves, per-class accuracy bar charts, confusion matrix heatmap, severity distribution, meter-position roll map, and conditional alert table — from the structured CSV outputs produced by the M3 batch pipeline. Dynamic slicers by Zone, Line, Defect Category, and Severity enable supervisors to drill into any cross-section of quality data.

The YOLOv8n classification model, achieving **93.48% Top-1 accuracy** and **100% Top-5 accuracy** on 552 validation images, forms a strong foundation for deployment. The primary improvement area is the Hole class (79% accuracy, confused with Stain), which will be addressed through targeted data augmentation and retraining.

Next Steps — Post Milestone 4:

- Retrain with augmented Hole and Vertical class data targeting Top-1 $\geq 95\%$
- Extend classification head to cover all 20 YOLOv11 detection classes
- Integrate real-time dashboard feed via MQTT/WebSocket from inference pipeline
- Deploy live system on NVIDIA GPU server with shop-floor SCADA alert integration
- Configure scheduled shift-summary PDF auto-export from dashboard
- Upload all M4 artefacts to GitHub repository